TECHNICAL PROJECT REPORT

Offline LoRa SOS Emergency System

ESP32 + LoRa SX1278 Mesh Network with Captive Portal & Rescue Dashboard

NODE 1
Transmitter

NODE 3
Relay

NODE 2
Base Station

Technology: ESP32 WROOM-32 · LoRa SX1278 Ra-02 · DHT12 Sensor

Frequency: 433 MHz · SF10 · 17 dBm · 125 kHz BW

Date: July 2026

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1 Project Overview & Objectives

This project implements an **offline, infrastructure-free SOS emergency communication system** using ESP32 microcontrollers and LoRa (Long Range) radio modules. The system is designed to operate in disaster scenarios such as earthquakes, floods, or remote area emergencies where cellular networks and internet connectivity may be unavailable.

A user in distress connects to a local WiFi access point created by Node 1 (Transmitter). A web-based SOS portal automatically appears on their phone. Upon triggering SOS, the device's GPS coordinates along with real-time temperature and humidity readings are broadcast via LoRa radio to Node 2 (Base Station). Rescue operators at Node 2 view live alerts on a web dashboard and dispatch acknowledgements back to the user.

Key Objectives

- Fully offline operation — no internet, no cellular, no infrastructure required
- One-click SOS with GPS coordinates captured from the user's phone browser
- Real-time temperature and humidity data sent with every SOS packet

- Automatic receive confirmation (user knows their SOS was received)
- Rescue team acknowledgement with status feedback to the user
- Long range coverage via LoRa mesh relay (Node 3)

2 System Architecture

The system consists of three ESP32 nodes communicating over LoRa radio at 433 MHz.

> **Node 1 (Transmitter)** → Creates WiFi AP " `SOS_EMERGENCY_AP` ". Hosts a captive portal web app. Reads GPS from phone browser. Reads DHT12 sensor. Broadcasts SOS via LoRa. Listens for RCV and ACK confirmations.
>
> **Node 3 (Relay — optional)** → Listens for all packet types and re-broadcasts them with hop count. Bridges the gap when Node 1 and Node 2 are out of direct range.
>
> **Node 2 (Base Station)** → Receives SOS packets. Displays alerts on a web dashboard at " `RESCUE_BASE_AP` " / `http://rescue.local` . Shows temperature & humidity. Auto-sends RCV confirmation. Operator clicks Resolve to send ACK.

```
[User's Phone]
     |  connects to SOS_EMERGENCY_AP
     ▼
[Node 1 — ESP32 + LoRa + DHT12]
     |  SOS:ID:LAT:LON:TEMP:HUMI  (433 MHz LoRa, SF10, 17 dBm)
     ▼
[Node 3 — Relay — optional]
     |  SOS:ID:LAT:LON:TEMP:HUMI:1  (hop count added)
     ▼
[Node 2 — ESP32 + LoRa + LCD]
     |  RCV:ID  ←——— auto confirmation
     |  ACK:ID  ←——— operator resolves
     ▼
[Rescue Operator's Device]
     connects to RESCUE_BASE_AP → views dashboard
```

3 Hardware Components

ESP32 WROOM-32 (30-pin)

Dual-core 240 MHz processor with built-in WiFi & Bluetooth. Used for all 3 nodes. RTOS support for concurrent WiFi and LoRa tasks.

LoRa SX1278 Ra-02 Module

433 MHz LoRa transceiver. SPI interface. 17 dBm output power. SF10 spreading factor for

long range. Capable of km-range communication.

DHT12 Sensor (Node 1)

Temperature and humidity sensor. Single-wire protocol. Connected to GPIO 15. Range: -20°C to 60°C temperature, 20–95% humidity.

16×2 I2C LCD (Node 2)

Displays incoming SOS alert coordinates and RSSI. Backlight flashes on new alert. Connected via I2C on GPIO 21 (SDA) and GPIO 22 (SCL).

100µF Capacitor

Placed across LoRa module VCC and GND. Absorbs current spikes during LoRa transmission at 17 dBm to prevent ESP32 brownout resets.

Built-in LED (Node 3)

GPIO 2 LED blinks once for each packet forwarded by the relay. 3 blinks on startup. Rapid blink indicates hardware error.

4 Hardware Wiring

LoRa SX1278 Module — All Nodes (Node 1, 2, 3)

LORA PIN	ESP32 GPIO	DESCRIPTION
VCC	3.3V	Power supply — 3.3V only (never 5V)
GND	GND	Ground — add 100µF cap here for stability
MISO	GPIO 19	SPI Master In Slave Out
MOSI	GPIO 23	SPI Master Out Slave In
SCK	GPIO 18	SPI Clock
NSS / CS	GPIO 5	SPI Chip Select
RST	GPIO 14	Reset Pin
DIO0	GPIO 4	Interrupt Pin (GPIO 4 avoids boot conflicts)

DHT12 Sensor — Node 1 Only

DHT12 PIN	ESP32 GPIO	DESCRIPTION
VCC	3.3V	Power supply
GND	GND	Ground
DATA	GPIO 15	Single-wire data — add 10kΩ pull-up to 3.3V

16×2 I2C LCD — Node 2 Only

LCD PIN	ESP32 GPIO	DESCRIPTION
VCC	5V (VIN)	Power supply — 5V for backlight
GND	GND	Ground
SDA	GPIO 21	I2C Data
SCL	GPIO 22	I2C Clock

> ⚠️ **Important:** The LoRa SX1278 operates strictly at **3.3V**. Connecting VCC to 5V will permanently damage the module. Always verify voltage before powering on.

5 LoRa Radio Configuration

All three nodes use identical radio parameters. Mismatched settings between nodes will cause packets to be unreadable.

PARAMETER	VALUE	PURPOSE
Frequency	433 MHz	ISM band — no license required
TX Power	17 dBm (50 mW)	Maximum safe power for SX1278 — long range
Spreading Factor	SF10	~5–10× more range than default SF7
Bandwidth	125 kHz	Standard — balance of range and speed
Coding Rate	4/8	Maximum error correction — recovers corrupt bits
CRC	Enabled	Rejects corrupted packets automatically

> **Range Improvement vs Original Settings:**
> Original code used TX Power = 2 dBm (1.5 mW) and SF7 — effective range ~50–100 metres. Updated settings (17 dBm + SF10) achieve **2–5 km** line-of-sight range with the SX1278. With Node 3 relay, total range extends to **4–10 km**.

6 Communication Protocol & Packet Format

All LoRa packets are plain ASCII text strings with colon-separated fields.

SOS Packet — Node 1 → Node 2

```
SOS:ID:LAT:LON:TEMP:HUMI
Example: SOS:1749000:12.97160:77.59460:28.5:62.0
```

Relayed SOS Packet — Node 3 adds hop count

```
SOS:ID:LAT:LON:TEMP:HUMI:HOP
Example: SOS:1749000:12.97160:77.59460:28.5:62.0:1
```

RCV Packet — Node 2 → Node 1 (auto-confirmation)

```
RCV:ID
Example: RCV:1749000
```

ACK Packet — Node 2 → Node 1 (operator resolved)

```
ACK:ID:HOP
Example: ACK:1749000:0
```

FIELD	DESCRIPTION
`ID`	Unique packet ID — milliseconds since boot (millis())
`LAT / LON`	GPS coordinates captured from user's phone browser
`TEMP`	Temperature in °C from DHT12 sensor (e.g. 28.5)
`HUMI`	Humidity percentage from DHT12 sensor (e.g. 62.0)
`HOP`	Relay hop count — incremented by each Node 3, max 3 hops

7 Node 1 — SOS Transmitter

Node 1 is the user-facing device carried by the person in distress.

Features

- Creates open WiFi Access Point: **SOS_EMERGENCY_AP**
- Serves a Captive Portal web app automatically when a phone connects
- Phone browser requests GPS location — coordinates displayed on screen
- DHT12 sensor reads temperature and humidity every 4 seconds
- SOS packet retransmitted every 4 seconds until RCV confirmation received
- Once RCV received — stops retransmitting (Node 2 already has the alert)
- Listens for ACK — updates status to SAFE when received
- 60-second timeout — resets to IDLE if no confirmation received
- RTOS dual-core: WiFi/HTTP on Core 1, LoRa TX/RX on Core 0 with SPI mutex

SOS Web Portal States



8 Node 2 — Base Station & Rescue Dashboard

Node 2 is operated by the rescue team. It receives SOS alerts and hosts a real-time web dashboard.

Features

- Receives SOS LoRa packets and parses all fields (ID, GPS, Temp, Humidity, RSSI)
- Automatically sends `RCV:<ID>` back to Node 1 on first receipt
- 16×2 LCD flashes backlight for 1 second on new alert, then stays solid ON
- LCD displays coordinates and RSSI signal strength
- Creates WiFi AP: **RESCUE_BASE_AP** — dashboard at `http://rescue.local` or `http://192.168.5.1`
- Dashboard shows all active and resolved alerts with GPS map links
- Each alert card shows: ID, coordinates, **temperature, humidity**, RSSI signal bars, distance/bearing from operator
- Operator clicks RESOLVE — Node 2 sends 3× ACK packets via RTOS background task
- Dashboard protected by 4-digit PIN lock
- WiFi station mode — can connect to router for internet access
- CSV export of all alert logs

Dashboard Alert Card (Updated)

> Each SOS alert card on the dashboard displays:
>
> 🔴 **SOS-001** *12s ago*
> Latitude: 12.97160 Longitude: 77.59460
> 🌡️ **28.5 °C** 💧 **62.0 %**
> 🧭 1.2 km away (NE) ▮▮▮▯ -78 dBm
> [📍 Map] [RESOLVE]

9 Node 3 — Mesh Relay

Node 3 extends the LoRa range by acting as a packet repeater between Node 1 and Node 2. It is only required when the two nodes are beyond direct LoRa range.

Relay Behaviour

PACKET TYPE	DIRECTION	ACTION
`SOS:…`	Node 1 → Node 2	Forward with incremented hop count

`RCV:…`	Node 2 → Node 1	Forward as-is
`ACK:…`	Node 2 → Node 1	Forward with incremented hop count

Anti-Loop Protections

- **Seen-Packet Cache:** Stores last 20 packet IDs for 60 seconds. Duplicate packets are silently dropped.
- **Hop Limit:** Maximum 3 hops. Packets with hop count ≥ 3 are dropped.
- **Random Jitter:** 50–300ms random delay before re-broadcasting — prevents collision when multiple relays exist.

> ✅ **Node 3 is optional.** If Node 1 and Node 2 are in direct range, Node 3 can be powered off. When powered on in range of both nodes, it transparently extends coverage with no configuration changes required.

10 DHT12 Temperature & Humidity Sensor

The DHT12 sensor on Node 1 provides environmental data that is included in every SOS transmission, helping rescue teams assess the survivor's environmental conditions.

PROPERTY	VALUE
Connected to	Node 1 — GPIO 15
Protocol	Single-wire (DHT12 compatible)
Library	Adafruit DHT sensor library (DHT12 type)
Temperature Range	-20°C to 60°C (±0.5°C accuracy)
Humidity Range	20% to 95% RH (±5% accuracy)
Pull-up Resistor	10kΩ between DATA and 3.3V

Sensor readings are taken before each LoRa transmission. If the sensor read fails (returns NaN), the packet uses `--` as placeholder. The dashboard hides the sensor row when no data is available.

11 SOS Status Flow

1. **User clicks SOS button**
   Phone sends GPS coordinates to Node 1 via HTTP. Status changes to SENDING. Button shows "BROADCASTING".
2. **Node 1 broadcasts SOS every 4 seconds**
   Reads DHT12 sensor. Transmits SOS:ID:LAT:LON:TEMP:HUMI via LoRa at 17 dBm, SF10.
3. **Node 3 relays (if needed)**
   Adds hop count, re-broadcasts with random jitter to Node 2.
4. **Node 2 receives SOS**
   Alert appears on dashboard with coordinates, temperature, humidity, RSSI. LCD flashes. Node 2 immediately sends RCV:ID back.
5. **Node 1 receives RCV**
   Status changes to RECEIVED. Node 1 stops retransmitting. User sees "RECEIVED BY RESCUE TEAM ✅".
6. **Operator resolves alert**
   Rescue operator clicks RESOLVE on dashboard. Node 2 broadcasts ACK:ID 3 times via RTOS background task.
7. **Node 1 receives ACK — User sees SAFE 🟢**
   Status changes to ACKNOWLEDGED. Button turns green showing "SAFE". Chime sound plays on phone.

12 Software Libraries

LIBRARY	AUTHOR	USED IN	PURPOSE
LoRa	Sandeep Mistry	All Nodes	LoRa packet TX/RX via SPI
WiFi	Espressif (built-in)	Node 1, 2	WiFi Access Point and Station mode
WebServer	Espressif (built-in)	Node 1, 2	HTTP server for web apps
DNSServer	Espressif (built-in)	Node 1	Captive portal DNS redirect

ESPmDNS	Espressif (built-in)	Node 2	rescue.local mDNS hostname
LiquidCrystal_I2C	Frank de Brabander	Node 2	16×2 I2C LCD control
DHT sensor library	Adafruit	Node 1	DHT12 temperature/humidity readings
Adafruit Unified Sensor	Adafruit	Node 1	Dependency for DHT library
FreeRTOS (built-in)	Espressif	Node 1, 2	Multi-core RTOS tasks with mutex

13 Testing Procedure

Step 1 — Flash Firmware

- Upload `node2_receiver.ino` to Node 2 ESP32
- Upload `node3_relay.ino` to Node 3 ESP32 (if using relay)
- Upload `node1_transmitter.ino` to Node 1 ESP32
- All at 115200 baud in Arduino IDE

Step 2 — Hardware Verification

- Open Serial Monitor (115200 baud) on Node 1 → confirm: `DHT12 sensor initialized` and `LoRa Active`
- Open Serial Monitor on Node 2 → confirm: `LoRa Initialization successful` and `Rescue Server started`
- Open Serial Monitor on Node 3 → confirm: 3 LED blinks and `LoRa Relay Active`

Step 3 — Mock SOS Test (No Phone Needed)

- In Node 2 Serial Monitor, type: `MOCK_SOS` and press Enter
- Connect to WiFi **RESCUE_BASE_AP** → open `http://192.168.5.1`
- Confirm alert appears with Temp: 28.5°C / Humi: 62.0%
- Click RESOLVE → confirm `ACK Sent` on Serial Monitor

Step 4 — Full System Test

- Connect phone to WiFi **SOS_EMERGENCY_AP**
- SOS portal opens automatically (or navigate to `http://192.168.4.1`)
- Allow GPS location access — coordinates appear on screen
- Click SOS button → confirm button shows "BROADCASTING"
- On Node 2 Serial Monitor: confirm coordinates and sensor data received
- On Node 2 dashboard: confirm alert card shows with temperature/humidity
- Click RESOLVE on dashboard → confirm phone shows "RECEIVED BY RESCUE TEAM ✅"
- After operator resolves → confirm phone shows "SAFE 🟢"

Step 5 — Range Test

- Start at 2 metres — confirm successful communication
- Move to 50 metres — confirm packet received (check Serial Monitor RSSI value)
- Move to 500 metres — confirm RSSI above -110 dBm
- If ESP32 crashes/resets during TX → add 100µF capacitor directly on LoRa VCC/GND

14 Conclusion

The Offline LoRa SOS Emergency System successfully demonstrates a fully self-contained, infrastructure-free emergency communication solution. The system operates entirely offline using local WiFi and LoRa radio, making it suitable for earthquake, flood, or remote area scenarios where conventional communication fails.

Key achievements of the project include: a captive portal web app requiring no app installation, real-time environmental sensing with the DHT12 sensor, a multi-state confirmation flow (BROADCASTING → RECEIVED → SAFE), a professional rescue operator dashboard with PIN security, and a mesh relay capability for extending range beyond 2 km.

The radio configuration upgrade from 2 dBm / SF7 to 17 dBm / SF10 increased effective range by approximately 10–20× compared to default settings, making the system viable for real-world emergency scenarios.

> ✅ **System is complete and ready for deployment.**
> Flash all 3 nodes, add 100µF capacitor on LoRa VCC/GND, power on, and the system operates fully autonomously with no internet connection required.